MIL-B-62098(AT)
18 June 1968

MILITARY SPECIFICATION

BULLDOZER, EARTH MOVING: TANK MOUNTING, M8A3

1. SCOPE

1.1 Scope. This specification covers hydraulically operated, M8A3 bulldozer kit, for mounting on the front of Tank, Combat, Full-Tracked: 90MM, M48A3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military
MIL-L-2104 - Lubricating Oil, Internal-Combustion Engine, Heavy Duty.
MIL-L-10295 - Lubricating Oil, Internal Combustion Engine, Sub-Zero.

STANDARDS
Military
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-193 - Painting Procedures Tactical Vehicles (Tracked and Wheeled).
MIL-STD-642 - Identification Marking Combat and Tactical Transport Vehicles.

PUBLICATIONS
Engineering Parts List
EPL 8736736 - Kit, Bulldozer Earth Moving M8A3.
EPL 10944880 - Kit, Installation for Kit Bulldozer M8A3.

FSC-2590

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SUPPLEMENTARY QUALITY ASSURANCE PROVISIONS

Master List

8736736 - Master List of Supplementary Quality Assurance Provision List for Bulldozer M8A3.

Packaging and Packing

8736736 - Packing Data Sheet for Bulldozer Kit M83, from M48A3.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction bulldozer kit. When specified (see 6.2), a complete bulldozer kit shall be produced prior to the manufacture or fabrication of kits in quantity. The complete kit shall be submitted to the Government for inspection to determine conformance to the requirements of this specification. Kit submitted by the contractor shall be fully representative of kits proposed to be supplied by the kit manufacturer from production facilities and tooling as contemplated under the supply contract. Bulldozer kit submitted shall be mounted on vehicle and operated through all ranges of operation.

3.2 First production kit. A first production kit fully representative of kits proposed to be furnished under the contract, shall be submitted to the Government for inspection to determine conformance to the requirements of this specification. The bulldozer kit when mounted on the M48A3 vehicle, shall be operated for a period of 10 hours, to assure that all performance requirements are met.

3.3 Materials. Materials shall be as specified on applicable drawings, specifications, and standards listed or referred to in Engineering Parts Lists (EPL) 8736736.

3.3.1 Qualified products. With respect to all assemblies and component parts requiring product qualification in accordance with the pertinent specifications or drawings listed on the EPL cited in 3.4, and necessary in the construction of this vehicle, the contractor shall be responsible for using only those products which are listed (by part or drawing number as QPL items) on the QPL in the referenced specification or which have been approved for inclusion on such lists. Manufacturer's inspection records shall specifically list such component(s), name of supplier(s) (as listed on the QPL or in the approval letter), and number and date of the QPL from which the selection was made or date of approval letter when products are approved but not as yet listed on the QPL.

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3.4 Construction. The bulldozer kits shall be constructed in accordance with drawings, specifications, and standards listed or referred to in EPL 8736736. All parts requiring identification shall be marked in accordance with MIL-STD-130.

3.4.1 Kit installation. Kit conforming to Drawing 8736736 shall be installed on the M48A3 vehicle in accordance with Drawing 10944880. Unless otherwise specified (see 6.2), one installation kit shall be furnished for each 10 bulldozer kits.

3.4.2 Service products. Bulldozer performance shall be demonstrated using standard service products, in the hydraulic system, of the following type and grade for the applicable ambient air temperature; at ambient air temperature from minus 10°F to plus 115°F, oil shall conform to grade 10 of MIL-L-2104; at ambient air temperature from minus 10°F to minus 40°F, oil shall conform to MIL-L-10295.

3.4.3 Hydraulic system. Provisions shall be made prior to initial filling to assure that oil confining areas of the hydraulic system are free of rust, scale, welding slag and spatter, or other contamination that may cause equipment failure.

3.5 Performance. The bulldozer, when using oil specified herein, completely assembled, and installed on the tank shall meet the performance requirements specified herein, and shall operate satisfactorily throughout all ranges of operation in the ambient temperature range of minus 40°F to plus 115°F.

3.5.1 Hydraulic operating system. The hydraulic system shall be powered by a hydraulic pump driven through a right angle drive speed reducer and magnetic clutch. A four-position control valve shall direct the oil flow to the double-acting hydraulic cylinders for control of the blade. The pump when operating at engine speed of 2800 rpm shall deliver a minimum flow of 60 gallons per minute at a pressure of 1000 psi.

3.5.1.1 Magnetic clutch. The magnetic clutch switch when in the on position shall cause the magnetic clutch to provide power for operation of the hydraulic pump.

3.5.2 Controls and linkage. All controls shall operate in the raise, lower, hold and float positions. The linkage shall operate in response to the controls throughout the full range of travel. The hydraulic system shall hold the moldboard at any height within operating range, and allow the moldboard to float at ground level.

3.5.3 Digging depth. Full downward pressure applied to piston shall hold the cutting edge of the blade to not less than 9 inches below ground level.

3.5.4 Rate of lift. The hydraulic system shall raise and lower the moldboard at the speeds listed in table 1. The specified rates shall be accomplished after the hydraulic system has been stabilized with an applied pressure of 1000 psi.

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Table I

Engine speed (RPM) MINIMUM	Minimum rate to raise and lower (inches per second)	
	Raise	Lower
1,000	2.5	6.5
1,500	4	6.5
2,000	5	6.5
2,400	5.5	6.5
2,800	6	6.5

3.6 Painting and marking.

3.6.1 Painting. Preparation for treatment and painting shall be in accordance with MIL-STD-193.

3.6.2 Marking. When specified (see 6.2), the marking of bulldozer shall be as directed by the procuring activity and in accordance with MIL-STD-642.

3.6.3 Nameplate. A nameplate conforming to applicable drawings shall be installed on the kit.

3.7 Technical literature. One copy of an installation instructions Drawing 11637638 a detail of EPL 8736736, shall be provided with each bulldozer kit.

3.8 Interchangeability. The bulldozer and components shall be constructed and assembled so that any part may be replaced with service parts, without modification.

3.9 Workmanship. Workmanship shall be of a quality to assure the kit and components free of defects resulting from improper manufacturing or assembly practices and shall conform to requirements herein specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Contractor's quality control system. Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective Quality Control System acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the Inspection station or to the method of Inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree or control required by this specification or other applicable documents referenced therein.

4.1.2 Parts and components. Parts, components and assemblies shall be inspected in accordance with the drawings listed on applicable EPL 8736736 and Master List of Supplementary Quality Assurance Provisions (SQAP) 8736736.

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveilliance of the operation to determine that practices, methods, and procedures of the written inspection plan are properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Qualified products. The contractor's inspection records shall list all components, name of suppliers, the number and date of applicable qualified products lists (see 3.3.1).

4.2 Preproduction and first production inspection.

4.2.1 Preproduction inspection. To determine conformance to 3.1 a preproduction bulldozer kit when furnished shall be inspected to ascertain kit conformance with the requirements of this specification. Installation and inspection of the bulldozer kit shall be performed at the location specified by the procuring activity (see 6.2). Failure of the preproduction model to comply with any of the requirements specified herein shall be cause for refusal of the Government to conduct a retest until corrective reserves satisfactory to the Government have been taken.

4.2.2 First production inspection.

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4.2.2.1 Special process inspection. A special process inspection will be conducted by the Government representatives during fabrication of the first production kit (see 3.2) to evaluate conformance of materials and workmanship with drawings and specification requirements.

4.2.2.1.1 Review and evaluation. Review and evaluation of records for processing procedure, inspection system and plan of quality control will be conducted on the following: material, surface hardening, heat treating, protective finishes, etc.

4.2.2.1.2 Bulldozer inspection. The bulldozer shall be inspected prior to covering the bulldozer with paint or other pertinents components or assemblies.

4.2.2.2 Complete bulldozer kit inspection. The first bulldozer inspection shall consist of an evaluation of workmanship and materials used by the contractor in the manufacturing of the kits. Evaluation shall be conducted by the Government at locations specified by the procuring activity. The kit shall be subject to examinations and checks as specified herein then installed on a vehicle and tested as specified in 3.2 and 4.4.

4.2.2.3 Test failure. Failure of the bulldozer kit to comply with any of the requirements specified or any deficiency of a workmanship or materials found on the kit during or as a result of the first production inspection, shall be cause for rejection of the kit. Further, the Government will refuse to continue acceptance of production vehicles until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiency. In addition, correction of the deficiency must be accomplished by the manufacturer at no cost to the Government. Any deficiency found as a result of first production test shall be considered prime facie evidence that all kits already accepted prior to completion of the test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies shall be corrected by the contractor at no cost to the Government.

4.3 Inspection provisions.

4.3.1 Examinations. Visual, dimensional, and primary functional examination of all completed kits shall be performed using the classification of defects shown in table II.

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Table II. Classification of Defects

Characteristics	Major	Minor	Method of Inspection
1. Right angle drive	Improper assy. clearance	Defective fittings, leaks appearance	Visual
2. Mounting brackets & mould board	Improper weld improper assy.	Adjustment	Visual
3. Hooks, lamps & guards	Improper assy. damage		Visual
4. Control assy.	Improper assy. damage, inoperative	Improper assy. & installation	Visual & functional
5. Installation fixture & templates	Damage		visual & functional
6. Hydraulic cylinder piping, guards	Leaks, improper assy. improper oil, foreign matter in system	Improper installation	Visual & functional
7. Painting	Inadequate coverage	Improper coverage application, materials	Visual
8. Identification & data plates		Incorrect data, size or location	Visual
9. Technical literature		Missing or incomplete	Visual

4.3.2 Production kit checks. During production all kits shall be checked for completeness of components, assemblies and cleanliness of system prior to shipment or storage.

4.4 Performance tests. For preproduction and first production inspection the bulldozer kit selected shall be subjected to all the tests specified herein.

4.4.1 Hydraulic operating system test. To determine conformance to 3.5 and 3.5.1, the bulldozer kit selected shall be mounted on a vehicle. The kit using specified oil shall be operated throughout all ranges of operations, and checked for system operating pressure, and hydraulic pump flow rate, at the engine operating speed specified.

4.4.1.1 Magnetic clutch check. To determine conformance to 3.5.1.1 with the control panel magnetic switch "on" the hydraulic pump shall actuate.

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4.4.2 Control and linkage test. To determine conformance to 3.5.2, during operating the controls and linkage shall be checked for functional requirements.

4.4.3 Digging depth test. To determine conformance to 3.5.3, the bulldozer kit shall be operated as specified and checked for digging depth, hold position and floating at ground level.

4.4.4 Rate of lift test. To determine conformance to 3.5.4, when using oils for specified temperatures and at operating pressures, the rate of lift for the moldboard shall be checked for all engine speeds shown in table I.

4.4.5 Failure. Failure of kit to meet specified requirements shall be cause for rejection and refusal of the Government to accept any additional production kits, until contractor has provided evidence satisfactory to the Government that defects causing rejection have been corrected.

4.5 Temperature extreme test. To determine conformance to 3.5, the bulldozer shall be subjected to operation and performance test at temperatures of minus 40°F and plus 115°F at locations specified by the procuring activity.

4.6 Preservation, packaging and kit processing. Material and equipment shall be inspected prior to shipment to assure conformance to applicable requirements specified in 5.1.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery and storage of the M8A3 bulldozer shall be in accordance with Packaging and Packaging Data Sheet 8736736.

6. NOTES

6.1 Intended use. The bulldozer, when mounted on the applicable tank (see 3.2), is intended for use in assault on obstacles, reduction of bunkers, and emergency construction operations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) If preproduction model is required (see 3.1).
(c) Kit installation required (see 3.4.1).
(d) Markings as required (see 3.6.2).
(e) Location of tests if required (see 4.2.1).

6.3 Supersession date. This specification includes the requirements of Interim Purchase Description ATPD-2021 dated 31 October 1967.

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. 2590-A0079

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